

ARUN PRAKASH BAAG

Senior CAE Engineering Leader • SAFe® Product Owner/Product Manager (POPM) • Google Certified Project Manager • Data Analytics Professional

Leamington Spa, United Kingdom • +44 744 88 58 616 • arunprakashbaag@gmail.com

24+

YEARS EXPERIENCE

15

YEARS CAE LEADERSHIP

6

VEHICLE PROGRAMMES LED

2

CERTIFICATIONS: SAFE POPM & GOOGLE PM

PROFESSIONAL SUMMARY

Accomplished engineering leader with 24 years of progressive experience (since September 2002) spanning Computer-Aided Engineering (CAE) simulation, automotive product development, data analytics, and Agile product leadership. A founding member of Jaguar Land Rover's Closures CAE team (since 2012) and earlier a founding member of the JLR ODEC India centre (2008-2010), with a consistent record of delivering complex, multi-programme CAE objectives, including concurrently leading two full vehicle programmes (Range Rover & Range Rover Sport) with the resource strength of a single team.

Recently certified as a SAFe® Product Owner/Product Manager (POPM) and a Google Certified Project Manager, and an active member of the Scaled Agile community, extending a strong foundation in formal project management methodology, Lean-Agile delivery, backlog ownership, and cross-functional Agile Squad leadership built on live vehicle programmes. Combines deep technical mastery of Closures Durability, NVH, Pedestrian Safety, and Crashworthiness engineering with a Data Analyst Apprenticeship (Distinction) and hands-on expertise in Python automation, Machine Learning, and Tableau data visualisation. A Lean Six Sigma Green Belt practitioner focused on turning engineering functions into data-driven, agile, high-performing organisations.

Now seeking a senior CAE, engineering leadership, data analytics, or SAFe Product Owner/Product Manager position where this blend of technical depth, Agile leadership, and analytics expertise can drive organisational excellence.

CORE COMPETENCIES

Engineering Leadership & Programme Management: Engineering Leadership, Chapter Lead, Team Leadership & Team Building, Project Management (Google Certified), Programme Management, Multi-Programme Delivery, Cross-Functional Collaboration, Stakeholder Management, Resource Optimisation, Risk & Change Management, Strategic Planning, Budget & Schedule Management, Talent Acquisition & Development, Mentorship & Coaching

Agile & Product Leadership (SAFe®): SAFe® Product Owner/Product Manager (POPM), Scaled Agile Framework (SAFe), Agile/Scrum Delivery, Agile Squad Leadership, Product Backlog Management & Prioritisation, Sprint & PI Planning, Iterative Delivery, Servant Leadership, Continuous Improvement

CAE Engineering & Simulation: Computer-Aided Engineering (CAE), Finite Element Analysis (FEA), Closures Durability, NVH (Noise, Vibration & Harshness), Pedestrian Head Impact & Safety (ECE127, EuNCAP), Crashworthiness, Vibration Fatigue, Stiffness Analysis, Squeak & Rattle, Topology/Topography Optimisation, DOE, DVP Optimisation, DFMEA

Data Analytics & Automation: Data Analysis & Analytics, Data Visualisation & Dashboarding, Business Intelligence, Predictive Analytics, Machine Learning Fundamentals, Python (Automation & Scripting), Tableau, Data Governance & Security, KPI Reporting, Root Cause & Trend Analysis

Process Excellence & Quality: Lean Six Sigma (Green Belt), Quality Assurance, Process Development & Implementation, Standard Work & Checklists, Warranty & Field Failure Root Cause Analysis

Simulation Software & Tools: ABAQUS, ANSA, HyperMesh, LS-DYNA, OptiStruct, PRIMER

PROFESSIONAL EXPERIENCE

Jaguar Land Rover Limited

CAE Lead • Full-time

11/2011 – Present

Gaydon, UK

- ▶ **Programme Leadership:** Led two mainstream vehicle programmes (Range Rover & Range Rover Sport) concurrently, managing extensive CAE scopes (Closures Durability, Pedestrian Head Impact, NVH) with the resource strength of a single team.
- ▶ **Talent Development:** Mentored and developed every new team member as a 'live resource' on lead programmes, ensuring rapid confidence-building and consistent delivery despite team attrition.
- ▶ **Team Culture:** Built a high-performing 'one-team' spirit through strategic work rotation, encouraging collaboration, shared

value, and continuous skill development.

- ▶ **Talent Acquisition & Development Champion:** Championed and implemented the intern and apprentice programme for the team (a first for the group), personally mentoring students from world-renowned engineering colleges to develop numerous customised automation applications for the team. Regularly support interview panels for new hires.
- ▶ **Project & Risk Management:** Applying Google-certified project management methodology (scope, schedule, risk, and stakeholder management) to plan and control CAE workstreams across concurrent, multi-programme vehicle deliveries.
- ▶ **Process Innovation:** Applied Green Belt (Lean Six Sigma) methodology to develop a robust process for robustness and sensitivity analysis on pedestrian head impact, directly contributing to DVP optimisation and enhancing result interpretation.
- ▶ **Automation & Data Engineering:** Engineered a Python application that automates ECE127 and EuNCAP impact point setup for Pedestrian Head Impact CAE, streamlining workflows and significantly boosting analyst efficiency.
- ▶ **Cross-Company Collaboration:** Collaborated with ARUP on development of the Pedestrian Head Impact Result mapping tool in PRIMER, providing input that materially enhanced capability.
- ▶ **Design Optimisation:** Provided technical expertise and guidance on optimisation methods (topology, topography, non-linear DOE) to identify cost- and weight-efficient vehicle and body designs that meet performance targets.
- ▶ **AI/ML & Agile Delivery:** Leading the strategic initiative to integrate Machine Learning and AI into Closures Durability and Pedestrian Head Impact CAE; delivering as a core member of a cross-functional Agile Squad on the EV Defender programme using sprint-based, iterative planning, directly applying SAFe® Product Owner/Product Manager principles to prioritise CAE deliverables against business value.

Affluent Technologies

CAE Lead • Contractor

Gaydon, UK

- ▶ As a founding member of the Closures CAE team (integrated into the design team), actively contributed to its establishment and growth, mentoring team members and ensuring smooth induction and continuous development.
- ▶ Led new process development and feasibility studies for automotive closures analysis, introducing practical checklists and efficient methodologies that improved quality and accelerated learning for new team members.
- ▶ Actively collaborated with design and manufacturing teams to identify potential issues early and develop mitigation actions, improving overall product quality.
- ▶ Piloted Closure Slam Analysis and Squeak & Rattle processes within the team, contributing to process development and test correlation.

Ford Technical Center India

Senior Specialist (CAE) – EASi Onsite Engineer • Full-time

09/2010 – 11/2011

India

- ▶ As Lead CAE Analyst at Ford Technical Center (client site), successfully restarted a critical Closures project from Ford Germany that had been halted for three years due to quality issues.
- ▶ Rebuilt client confidence through exceptional CAE work and reporting, receiving direct appreciation from both Ford Germany and Ford Australia; prepared a comprehensive guideline for future quality delivery, establishing new standards for project execution.
- ▶ Served as the CAE Durability Representative for EASi, actively engaging with clients for new project acquisition and providing support and mentorship to new engineers on Durability projects.

Tata Technologies

Senior CAE Analyst • Full-time

01/2007 – 09/2010

India

- ▶ Worked alone on GoodYear's Tire Project and subsequently drove formation of a full GoodYear ODEC centre within 3–4 months of joining.
- ▶ Led a CAE team within Tata Motors' Current Engineering Group, collaborating with a small integrated cross-functional team to resolve complex field failures. Resolved critical warranty issues for diverse automotive components (e.g. Knuckle Joints, BIW panels, Shock Tower cracks) using FEA/simulation, identifying root causes and recommending design changes to reduce claims and improve performance.
- ▶ Selected as a founding member of JLR ODEC India, leading knowledge transfer from Ford Technical Center and becoming the first member to establish and deliver all Closures Analysis disciplines for Evoque, significantly growing team confidence and stability.

HASETRI (R&D, J.K. Tyre)

Scientist C (CAE) • Full-time

01/2005 – 01/2007

India

- ▶ Conducted detailed footprint and deflection analyses to optimise tyre designs for commercial vehicles, contributing to improved safety, reliability, and efficiency.

- ▶ Taught and instructed students in various CAD and CAE software applications, building foundational knowledge in engineering design and analysis.

KEY VEHICLE PROGRAMMES

EV Defender (Current) — Agile Squad, CAE Lead

- ▶ Leading Bonnet & Fender Durability, NVH, and Pedestrian Head Impact as CAE Lead within a cross-functional Agile Squad, using Agile/sprint-based ways of working to deliver a new-feature Passive Bonnet under an Agile delivery system.
- ▶ Focused on continuous delivery of challenging requirements and meeting stakeholder needs with the highest priority, quality, and accuracy.

Range Rover & Range Rover Sport — CAE Lead

- ▶ Delivered two mainstream programmes concurrently with the resources of a single programme, showing multi-domain engineering and leadership capability.
- ▶ Pioneered the Strap design concept on Doors, leading to significant weight savings; developed new processes for Single Bolt hinge introduction and bolt pretension simulation, and established dynamic simulation processes for Power Door functionality.
- ▶ Delivered Pedestrian Head Impact analysis for both passive bonnets, achieving NVH- and Safety-optimised outcomes.

Range Rover Evoque — CAE Lead

- ▶ Directed Closures (Doors, Lids, Trims) and Pedestrian Head Impact CAE, stepping in midway through the programme cycle.
- ▶ Led Active Bonnet Pedestrian Head Impact CAE for the first time, mastering the attribute in parallel with delivery; introduced Noise Transfer Function (NTF) analysis to resolve persistent Tailgate Boom issues.

Discovery — CAE Lead

- ▶ Led Closures (Doors + Lids) Durability and NVH requirements; pioneered Laser Blank concept into the Front Door System, and resolved the most challenging aspect of the programme: optimising the heaviest rear tailgate.

Defender (Pre-PS Phase) — CAE Lead

- ▶ Led initial Pre-PS work for Doors, Lids, and Trims; supported foundational development of the taildoor concept enabling it to mount a 40kg tyre, establishing the pathway for advanced optimisation by subsequent teams.

DATA ANALYTICS & AUTOMATION PROJECTS

Data Champion, JLR Product Engineering

- ▶ Developed a full suite of customised Pedestrian Head Impact applications spanning Pre-Processing, Post-Processing, and Robustness functionality.
- ▶ **CAE Results Visualisation in Tableau:** Designed and implemented Tableau dashboards to deliver CAE results, improving data accessibility and visualisation for all stakeholders and turning complex datasets into actionable insight.
- ▶ **Closures Quality Data Visualisation Tool:** Collaborated with Quality Engineers to develop a specialised data visualisation tool for Closures quality data, featuring dynamic, interactive charts to track and analyse programme quality issues across JLR.

EDUCATION

Data Analyst Apprenticeship, Level 4

09/2022 – 02/2024*Institute for Apprenticeship & Technical Education — Distinction*

- ▶ Achieved Distinction while managing the curriculum alongside full-time CAE Lead responsibilities: stakeholder, project & risk management, data security, and data analytics.

Bachelor of Engineering

01/1998 – 01/2002*Utkal University, India*

Diploma in Business Administration

ICFAI University, India

CERTIFICATIONS

- ▶ SAFe® Product Owner/Product Manager (POPM) — Scaled Agile, Inc., 2026
- ▶ Member, SAFe® Community — Scaled Agile, Inc., 2026
- ▶ Google Project Management Certificate — Google, via Coursera, 2025
- ▶ Green Belt (Lean Six Sigma) — Bradford University & JLR Learning Academy, 02/2018 – 08/2018

AWARDS & LEADERSHIP BEYOND WORK

Professional and Technical Excellence Award

03/2025

JLR Learning Academy

JLR Hackathon Winner

03/2023

Jaguar Land Rover

- ▶ Led a team of four to win the JLR Hackathon, showing strong leadership and collaborative, innovative problem-solving.

Event Coordinator & Lead

01/2015 – 12/2023

Shree Krishna Mandir, Leamington Spa

- ▶ Led and coordinated numerous religious and cultural events, growing the organisation from a local entity into an established, recognised presence; built its website and grew fundraising, helping support new JLR engineers relocating from India. This growth led directly to new landmark premises in Leamington Spa.